

ASSIGNMENT 14

ROLL NO:CSB24083

NAME:DIPANKAR BEZBARUAH

```
import gc
import sys

class Node:
    def __init__(self, name):
        self.name = name
        self.link = None

    def __del__(self):
        print(f"{self.name} is being destroyed")

gc.disable()

print("===== Creating Nodes =====\n")

A = Node("Node A")
B = Node("Node B")

A.link = B
B.link = A

print("Cycle created:")
print("A -> B")
```

```
print("B -> A\n")

print("==== Reference Counts =====\n")

print("Reference count of A:", sys.getrefcount(A))
print("Reference count of B:", sys.getrefcount(B))

print("\n===== Deleting References =====\n")

del A
del B

print("Variables A and B deleted")
print("Objects now form an unreachable cycle\n")

print("===== Forcing Garbage Collection =====\n")

collected = gc.collect()

print("Unreachable objects collected:", collected)

print("\n===== Program Finished =====")
```

OUTPUT

```
===== Creating Nodes =====  
  
Cycle created:  
A -> B  
B -> A  
  
===== Reference Counts =====  
  
Reference count of A: 3  
Reference count of B: 3  
  
===== Deleting References =====  
  
Variables A and B deleted  
Objects now form an unreachable cycle  
  
===== Forcing Garbage Collection =====  
  
Node A is being destroyed  
Node B is being destroyed  
Unreachable objects collected: 2  
  
===== Program Finished =====
```